

Thamindu Weerasinghe

DATA SCIENCE UNDERGRADUATE | ASPIRING DATA ENGINEER

Colombo, Sri Lanka
+94765826516
nadeejatw@gmail.com

LinkedIn: <https://www.linkedin.com/in/thamindu-weerasinghe/>
GitHub: <https://github.com/Tnadeeja>
Portfolio: <https://thaminduweerasinghe-portfolio.vercel.app/>

Profile Summary

Data Science undergraduate at SLIIT with hands-on experience building ETL data pipelines, data warehouses, analytical dashboards, and backend systems through academic and team projects. Skilled in Python, SQL, data modeling, ETL development, data transformation, and dashboard development. Experienced in working with structured datasets, machine learning workflows, and REST API development. Seeking a Data Engineering internship to contribute skills in data pipelines, warehousing, and scalable data systems.

Technical Skills

- **Programming:** Python, Java, SQL, JavaScript, PHP
- **Data Engineering:** ETL, SSIS, Data Warehousing, Data Modeling, SCD Type 2, SSAS
- **Analytics & BI:** Power BI, Dashboarding, Statistical Analysis, Machine Learning
- **Databases:** MySQL, MongoDB, Oracle
- **Tools:** Git, GitHub, VS Code, RStudio
- **Web:** React, Node.js, Express.js, Next.js, HTML, CSS, Tailwind CSS

Project Experience

Data Warehouse & BI Project | ETL Pipeline, OLAP & Dashboard Development

SQL Server, SSIS, SSAS, Power BI

- Designed a star schema data warehouse with 1 fact table and 4 dimension tables for analytics reporting.
- Engineered ETL pipelines using SSIS to process data from 2 source files (CSV and TXT).
- Performed extraction, transformation, and loading across multiple structured datasets.
- Applied SCD Type 2 logic to maintain historical customer records.
- Created 1 OLAP cube in SSAS supporting drill-down and multidimensional analysis.
- Produced 4 Power BI reports covering sales trends, customer insights, products, and locations.

Machine Learning Analytics Project | Noise Impact on Model Robustness

Python, scikit-learn, React, Vite

- Built an end-to-end machine learning analytics project using the Red Wine Quality dataset (1,599 records, 11 features).
- Simulated 3 data quality issues: Gaussian noise, missing values, and outliers across multiple test scenarios.
- Trained and compared 3 regression models: Linear Regression, Random Forest, and Support Vector Regressor (SVR).
- Evaluated model robustness using RMSE, MAE, and R^2 under noisy conditions.
- Conducted statistical analysis using correlation profiling and one-way ANOVA to validate performance differences.
- Developed an interactive React dashboard with dynamic charts and filters to visualize model degradation trends.
- Automated result generation using Python pipelines for frontend-ready analytics reporting.

Backend Development Project | Smart Campus Operations Hub

Spring Boot, Java, React, MongoDB, GitHub

- Led development of the incident ticket management module within a 4-member Smart Campus platform supporting 3 user roles: Admin, Technician, and User.
- Implemented REST APIs for ticket creation, technician assignment, status transitions, comments, and attachment handling as part of a 53-endpoint system.
- Delivered workflow features including accept, reject, resolve, reassignment, and ownership-based edit/delete controls for role-specific operations.
- Architected MongoDB data models using 10 collections with validation logic to maintain ticket lifecycle consistency and secure access patterns.
- Championed the design of 35 Data Transfer Objects (DTOs) and 8 robust controllers, streamlining communication across the full-stack platform for improved efficiency.
- Collaborated in a 4-member team using GitHub branches, merge workflows, and module integration across bookings, facilities, notifications, and authentication.

Education

BSc (Hons) in Information Technology (Specializing in Data Science)

Sri Lanka Institute of Information Technology (SLIIT)

Sep 2023 – Present | Expected Graduation: 2027

GCE Advanced Level

Combined Mathematics | Physics | Chemistry

Harischandra National College, Negombo

Completed: 2021

Certifications

- Power BI Fundamentals - Microsoft
- Python Programming - University of Moratuwa
- AI/ML Engineer Stage 1 & 2 - SLIIT